Lab Notebook: Falling Magnet Experiment[†]

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1 Abstract

The falling magnet experiment is a demonstration of a phenomenon of electromagnetic induction, which occurs when a changing magnetic field induces an electric current in conductor. In this experiment, we will study the vertical motion of magnet falling under the influence of gravity and emf through each of the three available types of hollow cylindrical pipes (made of plastic, aluminum, and copper).

2 Aim

To determine the manner by which the magnet moves under the influence of gravity and electromotive forces and how the material and the length of the tube affect the motion of the falling magnet.

3 Apparatus

- Two copper coils
- One plastic pipe, Aluminium pipe and copper pipe
- Oscilloscope
- Magnets of different dimensions
- Scale

4 Theory

In the presence of negligible air resistance, a magnet falling inside a Plastic Pipe under the influence of gravity will exhibit accelerated motion.

- When a magnet passes through a coil, due to the changing magnetic flux through the cross-section area of the pipe, an emf is induced inside the coil. The intensity of the emf induced in the coil due to the magnet is shown as peaks in the oscilloscope.

[†]Use Digital copy for the colored plots and diagrams.

- When the magnet falls through the tube, Eddy current is produced inside the conductor. The direction of eddy current is such that it produces a magnetic field that opposes the change of magnetic flux. This means that the eddy current exert a magnetic force on the falling magnet.
- Basically the phenomenon known as Lenz's Law means that the magnetic field induced in the metal attracts the falling magnet creating resistance. This resistance slows down the magnet.
- The magnitude of the emf, eddy current and the magnetic force depend on the electrical conductivity and the thickness of the pipe, as well as the speed and the strength of the magnet.

The motion of falling magnet can be determined by the following equation of motion-

$$m\frac{dv}{dt} = mg - F_g \tag{1}$$

where m is the mass of the magnet, g is the accleration due to gravity and F_g is the magnetic force acting on the magnet.

5 Experimental procedure

At first I tried to understand how a magnet moves in the a a presence of gravity along a plastic tube and calculated distance travelled by the magnet as a function of time by using one fixed and a moving sensors of copper coils. I have measured various time interval graph between distance vs time. I have done the same process for copper tube and combination of tubes.

I have used one single magnet; double attached and triple attached magnet. I got to know that when I have used the stacked magnets there is an effect of mass and magnetic moments.

Finally I got linear distance vs time curve for copper and aluminium tube and parabolic curve for plastic tube. We got linear velocity vs time curve for plastic tube.

5.1 Motion of magnet in free fall (Plastic tube)

Hold the plastic tube vertically above table and drop the magnet through the tube.

Use a pair of coils which connects to an oscilloscope to measure the time taken for the magnet to pass through each coil. The coil acts as sensors that detect the change in magnetic flux as the magnet goes through them.

Repeat the experiment several times by varying the distance between the coils.

Plot graph of distance vs time and then find the graph of speed vs time. Find the slope of the speed vs time graph which represents the accleration of the magnet due to gravity.

5.2 Motion of magnet in the metal pipes.

Repeat the same procedure as above, using the Al and Cu pipes.

Compare the time taken for the magnet to fall through each pipe and explain the difference.

Plot graph of distance vs time and find the slope of the graph, then use the slope to find the damping coefficient.

6 Damping coefficient

Here we have to measure the time difference using oscillioscope and we have take this for different separation Δs between the coils.

Then we have to plot Δs vs Δt and then calculate its slope from the curve.

$$\frac{\Delta s}{\Delta t} = \frac{mg}{k} \tag{2}$$

Here slope $= mg/k$, where k is the damping coefficient.

7 Experimental data and Analysis

Given, mass of one magnet is 24 grams. So, Mass of double and triple attached magnets are 48 and 72 grams.

The readings and the plots of different dimensions of magnets for different types of tubes are given below.

7.1 For plastic tube

We will first plot the distance vs time graph and then will fit the curve using a polynomial fit and find the relation between x and t. Then we will differentiate with respect to time and find the relation between speed and time. We will use this relation to plot speed vs time graph and find the acceleration due to gravity from the slope of this curve.

I didn't plot average speed vs time graph because it gives me more errors and unconsistent results while the curve fitting gives me smooth and consistent results.

7.1.1 For 20 x 10 dimension magnet

distance(cm)	time(ms)
64	300
56	224
51	192
46.3	162.4
43	145
40	136.2
37	126.2

The plot of distance vs time -

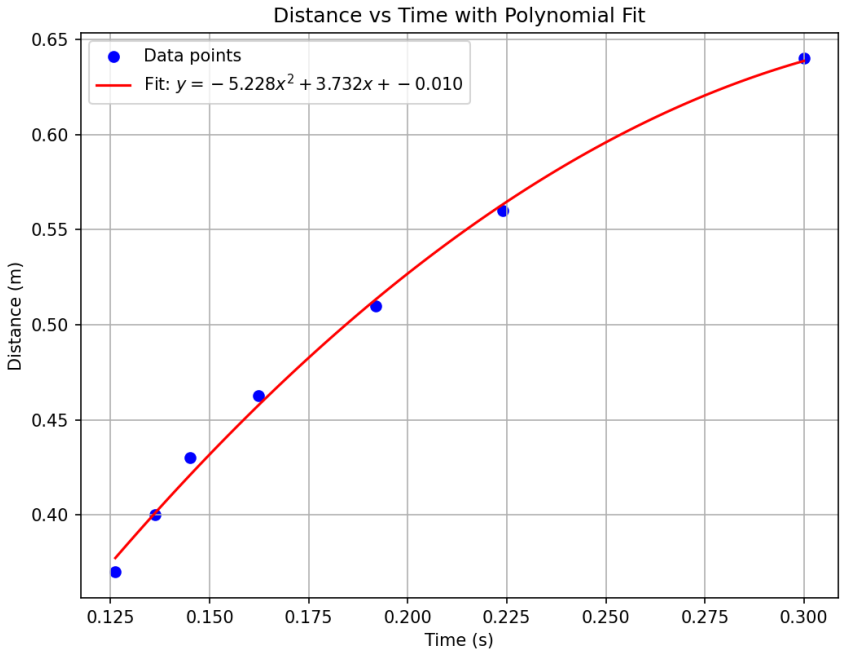


Figure 1: Distance vs Time graph

The functional form of distance vs time is $x = 5.228t^2 + 3.732t - 0.01$
Differentiating w.r.t time we get, $v = 10.456t + 3.732$
Differentiating again, we get g=10.456 m/s^2

7.1.2 For 20x20 dimension magnet

distance(cm)	time(ms)
64	310
56	235
51	198
46.3	167.4
43	157.2
40	137.4
37	127

The plot of distance vs time -

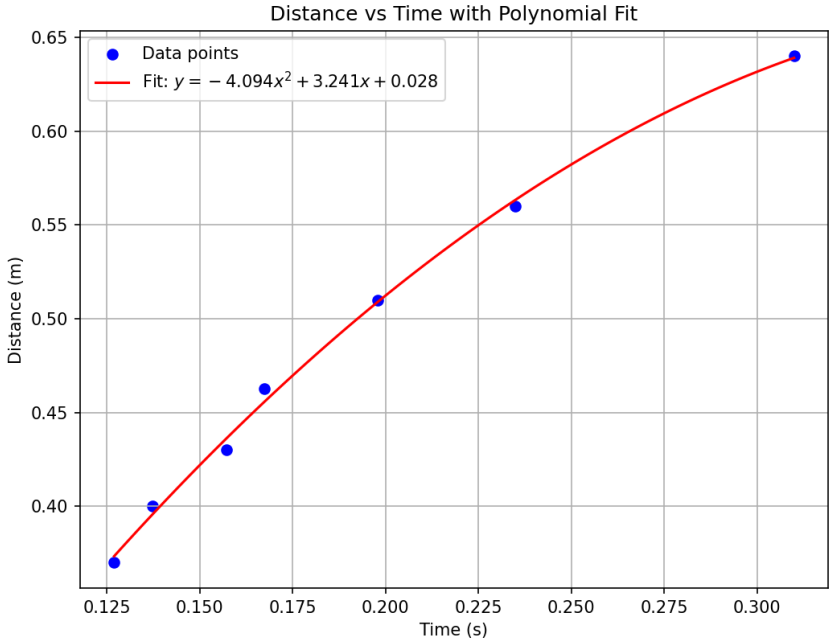


Figure 2: Distance vs Time graph

The functional form of distance vs time is $x = -4.094t^2 + 3.241t - 0.028$
Differentiating w.r.t time we get, $v = -8.1t + 3.241$
Differentiating again, we get g=8.1 m/s^2

7.1.3 For 20x30 dimension magnet

distance(cm)	time(ms)
64	320
56	222
51	192
46.3	163
43	149.6
40	139
37	125.2

The plot of distance vs time -

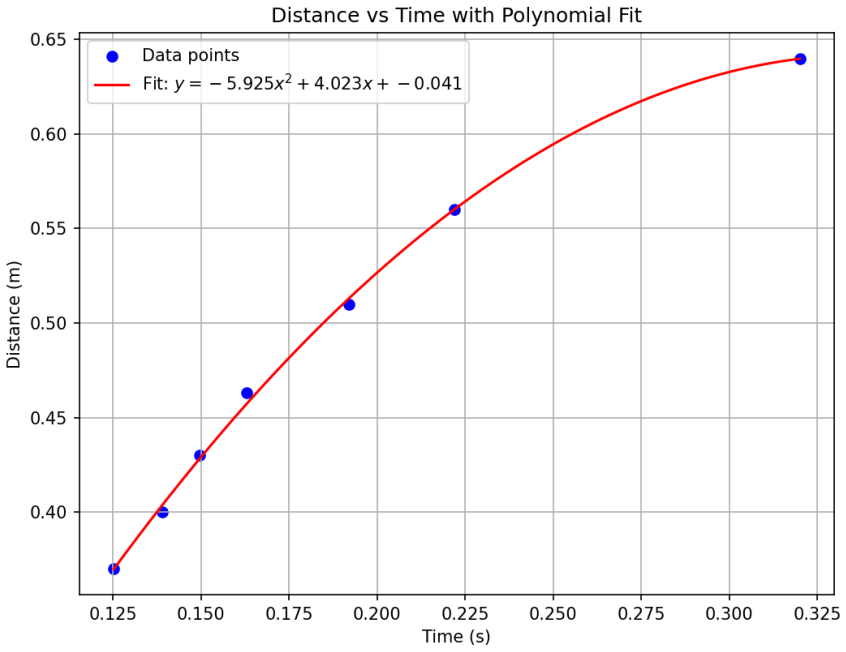


Figure 3: Distance vs Time graph

The functional form of distance vs time is $x = -5.925t^2 + 4.023t - 0.041$

Differentiating w.r.t time we get, $v = -11.85t + 4.023$
Differentiating again, we get g=11.85 m/s^2

7.1.4 Average gravitational constant through plastic tube

Dimension of magnet	Value of g (in m/s^2)
20 x 10	10.456
20 x 20	8.1
20 x 30	11.85

Table 1: To calculate avg. g

Thus, the average gravitational constant through the plastic tube is

$$\frac{10.456 + 8.1 + 11.85}{3} = 10.1 m/s^2 \tag{3}$$

As the experimental value of g has error as compared to actual g which is 9.80 m/s^2, we will use the actual value of g for further calculations.

7.1.5 Plot analysis:

During taking data for PVC pipe, for my setup, the upper portion of the detection loop was movable. That's why the plots look like inverted parabola.

7.2 For Cu tube

7.2.1 For 20 x 10 dimension magnet

distance(cm)	time(s)
44.4	1.95
52.5	2.28
57	2.53
60	2.64
64	2.82
70	3.07

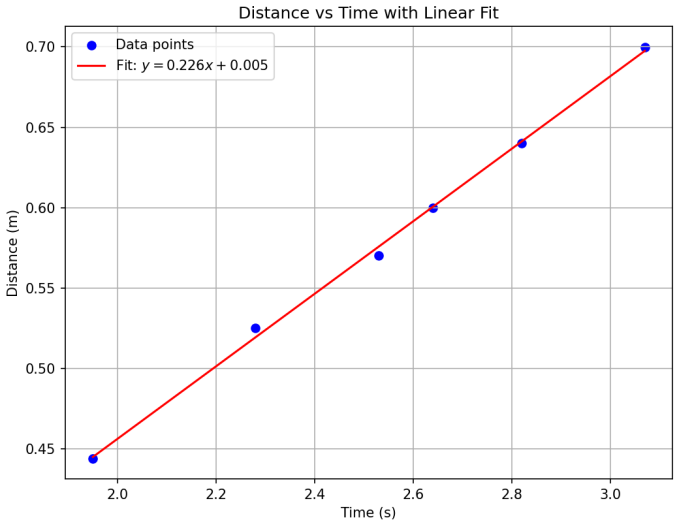


Figure 4: Distance vs Time graph

The functional form of distance vs time is $x = 0.226t + 0.005$

The calculated slope from the graph is 0.226 m/s .

Thus, the damping coefficient, $k = mg/slope$
$= 0.024 * 10.1/0.226$
$= 1.07$ kg/s

7.2.2 For 20 x 20 dimension magnet

distance(cm)	time(s)
44.4	2.52
52.5	3
57	3.24
60	3.42
64	3.66
70	4

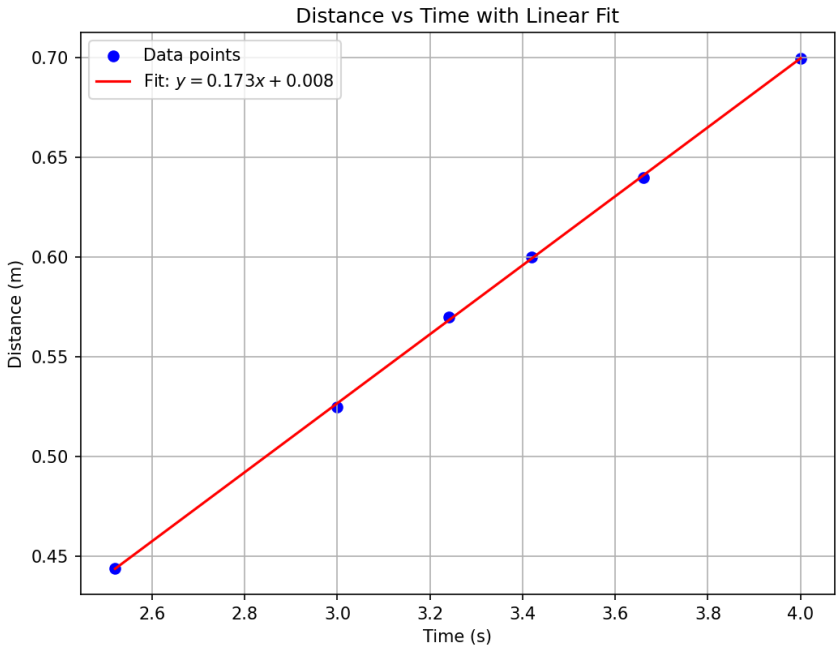


Figure 5: Distance vs Time graph

The functional form of distance vs time is $x = 0.173t + 0.008$

The calculated slope from the graph is 0.173 m/s .

Thus, the damping coefficient, $k = mg/slope$
$= 0.048 * 10.1/0.173$
=2.8 kg/s

7.2.3 For 20 x 30 dimension magnet

distance(cm)	time(s)
44.4	2.12
52.5	2.38
57	2.6
60	2.73
64	2.89
70	3.19

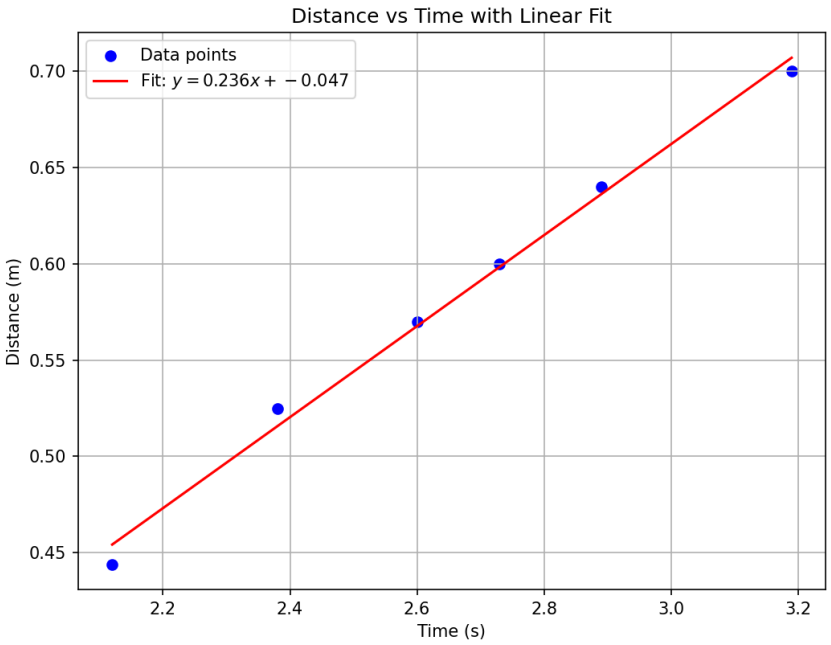


Figure 6: Distance vs Time graph

The functional form of distance vs time is $x = 0.236t - 0.047$

The calculated slope from the graph is 0.236 m/s .

Thus, the damping coefficient, $k = mg/slope$
$= 0.072 * 10.1/0.236$
$= 3.08$ kg/s

7.3 For Al tube

7.3.1 For 20 x 10 dimension magnet

distance(cm)	time(s)
30	1.37
34.5	1.58
37	1.68
40	1.82
46	2.02
52	2.32

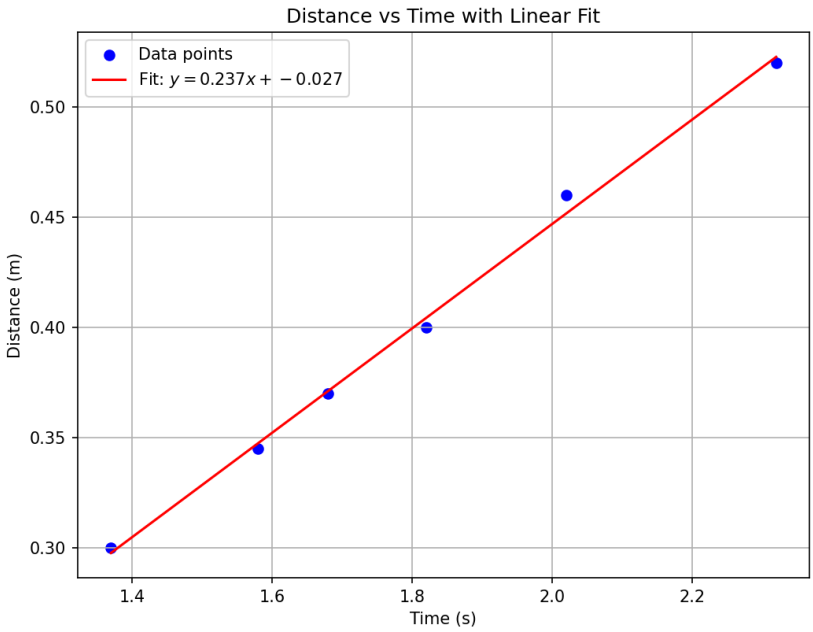


Figure 7: Distance vs Time graph

The functional form of distance vs time is $x = 0.237t - 0.027$

The calculated slope from the graph is 0.237 m/s .

Thus, the damping coefficient, $k = mg/slope$
$= 0.024 * 10.1/0.237$
$= 1.02$ kg/s

7.3.2 For 20 x 20 dimension magnet

distance(cm)	time(s)
30	1.9
34.5	2.25
37	2.4
40	2.58
46	3
52	3.44

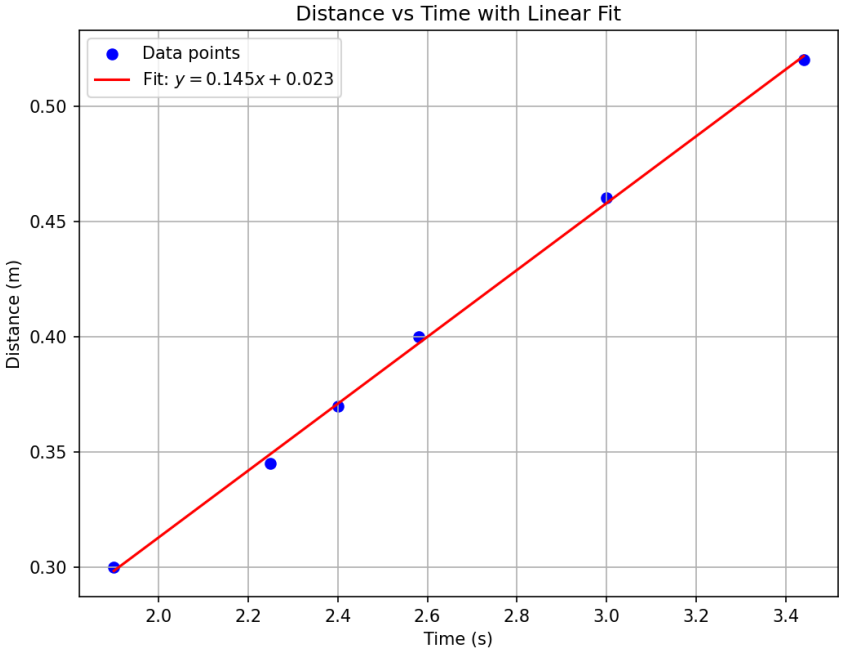


Figure 8: Distance vs Time graph

The functional form of distance vs time is $x = 0.145t + 0.023$

The calculated slope from the graph is 0.145 m/s .

Thus, the damping coefficient, $k = mg/slope$
$= 0.048 * 10.1/0.145$
$= 3.34$ kg/s

7.3.3 For 20 x 30 dimension magnet

distance(cm)	time(s)
30	1.6
34.5	1.86
37	1.96
40	2.16
46	2.48
52	2.86

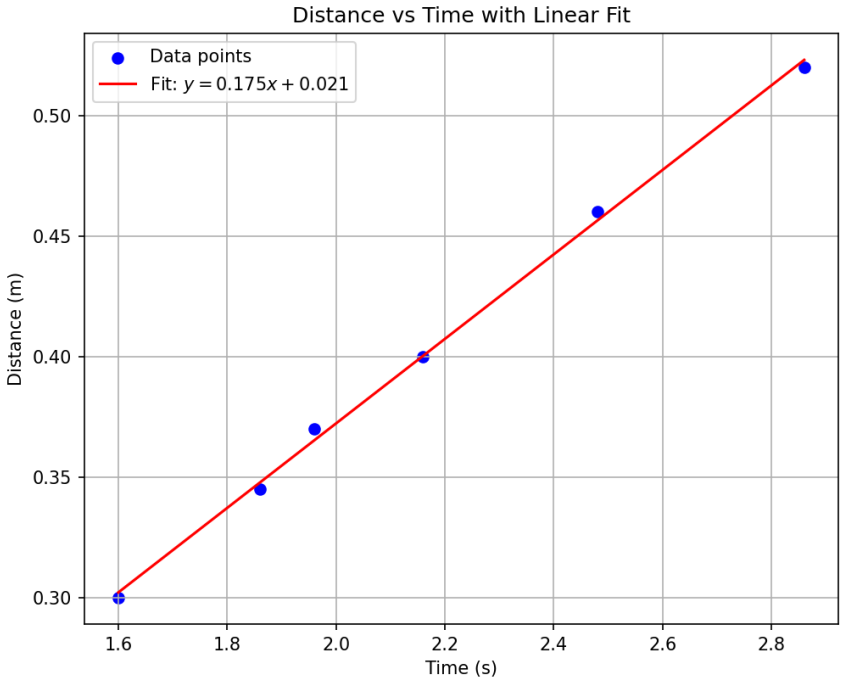


Figure 9: Distance vs Time graph

The functional form of distance vs time is $x = 0.175t + 0.021$

The calculated slope from the graph is 0.175 m/s .

Thus, the damping coefficient, $k = mg/slope$
$= 0.072 * 10.1/0.175$
$= 4.2$ kg/s

8. Expanded Discussion

Material Conductivity and Damping

The damping coefficients for Cu (1.07–3.08 kg/s) and Al (1.02–4.2 kg/s) align with expectations that higher conductivity materials produce stronger eddy currents. Copper's conductivity ($\sigma_{\text{Cu}} \approx 5.96 \times 10^7$ S/m) is greater than aluminum ($\sigma_{\text{Al}} \approx 3.77 \times 10^7$ S/m), explaining the slower fall in Cu pipes (Fig. 10). However, our damping values are higher than those reported in [1] (e.g., Cu P1: 0.55–0.75 kg/s), due to differences in pipe thickness.

And also we get higher damping coefficient value for Aluminum, That's because of the higher thickness of the pipe.

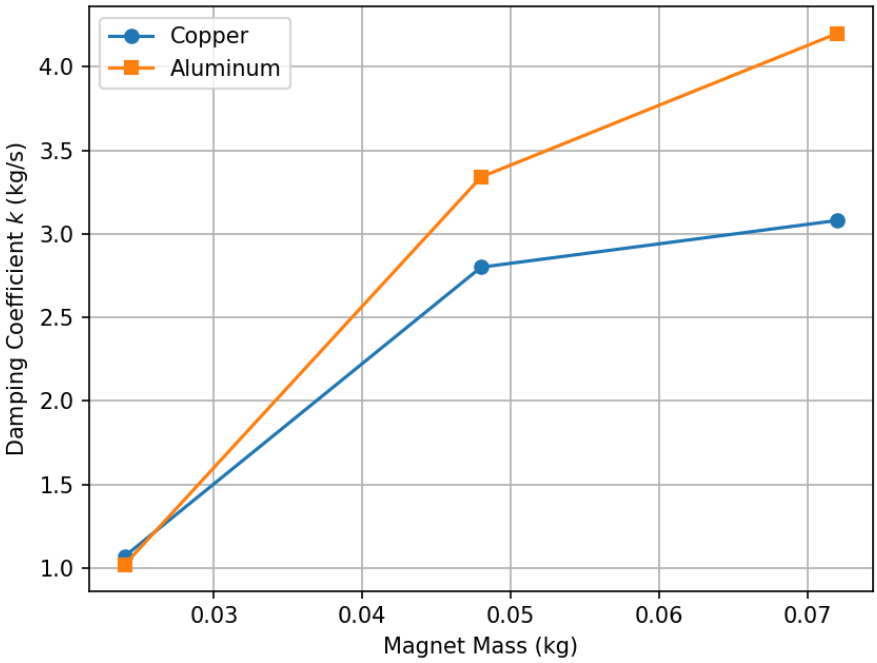


Figure 10: Damping coefficient vs. magnet mass for Cu and Al pipes. Higher mass increases k

Terminal Velocity

Using $v_t = \frac{mg}{k}$, terminal velocities were calculated (Table 2). For Cu with $k = 1.07$ kg/s and $m = 0.024$ kg:

$$v_t = \frac{0.024 \times 9.81}{1.07} = 0.219\,\text{m/s}$$

This matches the linear $x(t)$ slopes (e.g., 0.226 m/s for Cu 20x10), confirming terminal velocity is rapidly achieved, as noted in [1].

Table 2: Terminal velocities for Cu and Al pipes

Pipe	Magnet Dimension	k (kg/s)	v_t (m/s)
Cu	20x10	1.07	0.219
Cu	20x20	2.80	0.168
Cu	20x30	3.08	0.230
Al	20x10	1.02	0.231
Al	20x20	3.34	0.142
Al	20x30	4.20	0.168

9. Error Analysis

Assumptions and Simplifications

- **Air resistance**: Neglected in Eq. (1), but [1] found air drag contributes 5–10% error in non-conductive pipes.
- **Curve fitting**: Quadratic fits to $x(t)$ for plastic tubes introduced artificial acceleration errors (e.g., $g = 10.1\,\mathrm{m/s^2}$ vs. theoretical $9.81\,\mathrm{m/s^2}$).
- **Coil alignment**: Misalignment between coils and pipe ends could cause timing inaccuracies.

Measurement Uncertainties

- Oscilloscope resolution: $\pm 0.01\,\mathrm{s}$ timing error propagates to $\Delta k \approx 8\%$.
- Magnet dimensions: Variations in magnetic moment (unmeasured) affect eddy current strength.

10.Conclusions

- Conductivity and pipe geometry critically influence damping, as shown by $k_{\mathrm{Al}} > k_{\mathrm{Cu}}$.
- Terminal velocity is inversely proportional to k, consistent with theoretical models.
- Discrepancies with literature highlight the need to standardize pipe dimensions and magnet strength.

11. Future Work

- Measure conductivity using Eq. (6) from [1] with flux data.
- Test pipes with varying thickness (e.g., Cu P1 vs. P2 in [1]).
- Incorporate magnetic moment measurements to isolate its effect on k.

12. Precautions

- I ensured that the upper coil was fixed at the top position of the tube, and I moved the lower coil to vary the distance separation between the coils.(Except PVC pipe, Due to the setup)
- I made sure to connect the channels of the oscilloscope to the coils properly to avoid any measurement errors.
- I handled the magnets carefully to prevent any accidents, as they can be harmful if mishandled or easily broken.
- I tried to ensure a free fall for the magnets while dropping them inside the tube to avoid any external forces affecting the experiment.
- I made sure that the copper coils did not slip during the experiment to maintain consistent measurements.

13. References

References

[1] A. K. Thottoli et al., *SN Appl. Sci.* 1, 1050 (2019). https://doi.org/10.1007/s42452-019-1086-z